The primal-dual method

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The reference pair of primal and dual problems

Introduction: The reference pair of primal and dual problems

- The simplex algorithm and the dual simplex algorithm move from **basis** to basis: the **primal-dual method** is a third way to solve an LP problem
- We fix the pair of primal and dual problems on which the method works: a problem in **standard minimization form** and its dual
- Then we define the **admissible set** of a dual feasible solution, and we see how the Complementary Slackness theorem turns it into the idea of the method

Three ways to reach optimality

- The simplex algorithm keeps a **primal feasible** basis and works towards dual feasibility
- The dual simplex algorithm keeps a **dual feasible** basis and works towards primal feasibility
- The **primal-dual method** does something different:
  - it keeps a dual feasible solution, which is not required to be a basic one
  - it uses it to decide which primal variables are allowed to take a positive value

The reference pair of primal and dual problems

The method is naturally stated for an LP problem in **standard minimization form** with non-negative right-hand side, and for its dual:

Primal problem

$$\begin{aligned} \min \quad & \boldsymbol{c}' \boldsymbol{x} \\ \text{subject to} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}, \\ & \boldsymbol{x} \geq 0 \end{aligned}$$

Dual problem

$$\begin{aligned} \max \quad & \boldsymbol{b}' \boldsymbol{\pi} \\ \text{subject to} \quad & \boldsymbol{A}' \boldsymbol{\pi} \leq \boldsymbol{c}, \\ & \boldsymbol{\pi} \gtreqless 0 \end{aligned}$$

with $\boldsymbol{b} \geq 0$.

Reading the pair

- The n dual constraints $\boldsymbol{A}' \boldsymbol{\pi} \leq \boldsymbol{c}$ read

$$\boldsymbol{A}'_j \boldsymbol{\pi} \leq c_j, \qquad \forall j \in \{1, 2, \ldots, n\},$$

  where $\boldsymbol{A}_j$ is the j-th column of the constraint matrix

- The m dual variables are free, since the primal constraints $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$ are equalities

Every LP problem can be brought to this form

- A maximization objective function is replaced by the opposite one
- The inequality constraints receive slack or surplus variables
- The non-positive and free variables are replaced as usual
- Every constraint with a negative right-hand side is multiplied by -1

Given a dual feasible solution, some dual constraints are satisfied with equality:

their indices form the **admissible set**.

Definition 1: admissible set

Given a feasible solution $\boldsymbol{\pi}$ of the dual problem, the ***admissible set*** *is the set of the indices of the dual constraints satisfied with equality:*

$$\mathcal{J}(\boldsymbol{\pi}) = \left\{ j \in \{1, 2, \ldots, n\} \ : \ \boldsymbol{A}_j' \boldsymbol{\pi} = c_j \right\}.$$

Complementary Slackness and the admissible set

- By the Complementary Slackness theorem, a feasible solution $\boldsymbol{x}$ of the primal problem and a feasible solution $\boldsymbol{\pi}$ of the dual problem are both optimal if and only if

$$x_j = 0, \qquad \forall j \notin \mathcal{J}(\boldsymbol{\pi}),$$

- that is, if and only if only the admissible variables take a positive value

The idea of the method

- Given a dual feasible π, we look for a primal feasible solution which uses only the variables of $\mathcal{J}(\pi)$
- If such a solution exists, the pair is **optimal**
- If it does not, the attempt itself tells us how to improve π

Summary: the reference pair of primal and dual problems

The pair:

- Primal: min $\boldsymbol{c}'\boldsymbol{x}$ subject to $\boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$, $\boldsymbol{x} \geq 0$, with $\boldsymbol{b} \geq 0$
- Dual: max $\boldsymbol{b}'\boldsymbol{\pi}$ subject to $\boldsymbol{A}'\boldsymbol{\pi} \leq \boldsymbol{c}$, with $\boldsymbol{\pi}$ free

The admissible set:

- $\mathcal{J}(\boldsymbol{\pi})$: the indices of the dual constraints satisfied with equality
- The pair is optimal if and only if only the admissible variables take a positive value

The restricted primal problem

Introduction: The restricted primal problem

- To check whether the admissible variables are enough, we solve an auxiliary LP problem: the **restricted primal problem**
- Its optimal value tells whether the current pair is optimal
- Its dual, the **restricted dual problem**, gives the **direction** along which the dual solution is improved

Using only the admissible variables

- Given a dual feasible solution π with admissible set $\mathcal{J}$, we look for a primal feasible solution using only the variables with index in $\mathcal{J}$
- This is done by solving an auxiliary problem with m **artificial variables**, exactly as in the Phase 0 of the simplex algorithm

Definition 2: restricted primal problem

Given a feasible solution π of the dual problem with admissible set $\mathcal{J}$, the ***restricted primal problem*** *$RP(\mathcal{J})$ is:*

$$\xi(\mathcal{J}) = \min \sum_{i=1}^{m} a_i$$

$$\text{subject to} \quad \sum_{j \in \mathcal{J}} \boldsymbol{A}_j x_j + \boldsymbol{a} = \boldsymbol{b},$$

$$x_j \geq 0, \quad \forall j \in \mathcal{J},$$

$$\boldsymbol{a} \geq 0.$$

The restricted primal problem is always solvable

- It is **never infeasible**: since $\boldsymbol{b} \geq 0$, it has the basic feasible solution

$$\boldsymbol{a} = \boldsymbol{b} \qquad \text{and} \qquad x_j = 0, \quad \forall j \in \mathcal{J}$$

- It is **never unbounded**: its objective value is non-negative

The two outcomes of the restricted primal problem

1. If $\xi(\mathcal{J}) = 0$, then the optimal solution of $RP(\mathcal{J})$, completed with $x_j = 0$ for every $j \notin \mathcal{J}$, is a **feasible solution of the primal problem** satisfying the Complementary Slackness conditions together with $\boldsymbol{\pi}$: the pair is **optimal** and the method stops

2. If $\xi(\mathcal{J}) > 0$, then no primal feasible solution uses only the admissible variables, and the dual solution $\boldsymbol{\pi}$ has to be improved

To improve π we look at the dual
of the restricted primal problem.

Definition 3: restricted dual problem

The dual problem of the restricted primal problem $RP(\mathcal{J})$ *is called the* ***restricted dual problem*** $DRP(\mathcal{J})$*:*

$$\max \quad \boldsymbol{b}' \boldsymbol{y}$$

$$\text{subject to} \quad \boldsymbol{A}'_j \boldsymbol{y} \leq 0, \quad \forall j \in \mathcal{J},$$

$$y_i \leq 1, \quad \forall i \in \{1, 2, \ldots, m\}.$$

The optimal solution of the restricted dual problem

- Its optimal solution $\boldsymbol{y}$ is produced by the simplex algorithm together with the optimal solution of $RP(\mathcal{J})$
- By Strong Duality we have

$$\boldsymbol{b}'\boldsymbol{y} = \xi(\mathcal{J}) > 0$$

The improving direction

- The vector $\boldsymbol{y}$ is the **direction** along which the dual solution is improved:

$$\boldsymbol{\pi} \longleftarrow \boldsymbol{\pi} + \theta\, \boldsymbol{y}$$

- The dual objective value increases, for every $\theta > 0$, by

$$\boldsymbol{b}'(\boldsymbol{\pi} + \theta\, \boldsymbol{y}) - \boldsymbol{b}'\boldsymbol{\pi} = \theta \underbrace{\boldsymbol{b}'\boldsymbol{y}}_{=\xi(\mathcal{J})>0} > 0$$

Summary: the restricted primal problem

Restricted primal problem $RP(\mathcal{J})$:

- Only the admissible variables and m artificial variables; it minimizes the sum of the artificial variables
- Never infeasible, never unbounded
- $\xi(\mathcal{J}) = 0$: the pair is optimal; $\xi(\mathcal{J}) > 0$: $\boldsymbol{\pi}$ has to be improved

Restricted dual problem $DRP(\mathcal{J})$:

- Its optimal solution $\boldsymbol{y}$ has $\boldsymbol{b}'\boldsymbol{y} = \xi(\mathcal{J}) > 0$: it is an improving direction for $\boldsymbol{\pi}$

The dual step

Introduction: The dual step

- Along the direction $\boldsymbol{y}$ the dual objective value increases: how far can we move?
- We compute the **step length** θ, the largest one that keeps the dual solution feasible, and we see when θ can grow without bound
- Then we see how the **admissible set** changes after the step, and how to find an initial dual feasible solution

Keeping the dual solution feasible

- The step length θ is the largest value for which the new dual solution remains feasible
- For every $j \in \{1, 2, \ldots, n\}$ we need

$$\boldsymbol{A}_j'(\boldsymbol{\pi} + \theta\, \boldsymbol{y}) \leq c_j \qquad \Longleftrightarrow \qquad \theta\, \boldsymbol{A}_j' \boldsymbol{y} \;\leq\; \underbrace{c_j - \boldsymbol{A}_j' \boldsymbol{\pi}}_{\text{dual slack } s_j \geq 0}$$

Which constraints limit the step

- For $j \in \mathcal{J}$ the constraint is satisfied for every $\theta \geq 0$, since

$$s_j = 0 \qquad \text{and} \qquad \boldsymbol{A}_j' \boldsymbol{y} \leq 0$$

  by definition of $DRP(\mathcal{J})$

- For $j \notin \mathcal{J}$ we have $s_j > 0$, and the constraint restricts θ only when

$$\boldsymbol{A}_j' \boldsymbol{y} > 0$$

The step length

The **step length** is

$$\theta = \min \left\{ \frac{s_j}{\boldsymbol{A}_j' \boldsymbol{y}} \ : \ j \notin \mathcal{J}, \ \boldsymbol{A}_j' \boldsymbol{y} > 0 \right\}, \qquad \text{with} \qquad s_j = c_j - \boldsymbol{A}_j' \boldsymbol{\pi}.$$

- If $\boldsymbol{A}_j' \boldsymbol{y} \leq 0$ for every $j \notin \mathcal{J}$, then θ can grow without bound: the dual problem is **unbounded**
- By the theorem on the possible outcomes of a pair of primal and dual problems, the primal problem is then **infeasible**

After the step, some indices enter the admissible set and some may leave it.

How the admissible set changes

- The index $j^\star$ attaining the minimum **enters** the admissible set: after the step, its dual slack becomes zero
- Every index j of a basic variable of the optimal solution of $RP(\mathcal{J})$ **remains** in the admissible set, since $\boldsymbol{A}'_j \boldsymbol{y} = 0$ by the Complementary Slackness conditions between $RP(\mathcal{J})$ and $DRP(\mathcal{J})$
- Only the indices $j \in \mathcal{J}$ with $\boldsymbol{A}'_j \boldsymbol{y} < 0$ may **leave** the admissible set, since their dual slack becomes

$$-\theta \, \boldsymbol{A}'_j \boldsymbol{y} > 0$$

The initial dual feasible solution

- The initial dual feasible solution is often immediate
- If $\boldsymbol{c} \geq 0$, then $\boldsymbol{\pi} = 0$ is feasible, with

$$\mathcal{J} = \{ j : c_j = 0 \}$$

- Otherwise, a dual feasible solution has to be determined beforehand

Summary: the dual step

Step length:

- $\theta = \min\{s_j / \boldsymbol{A}'_j \boldsymbol{y} : j \notin \mathcal{J},\ \boldsymbol{A}'_j \boldsymbol{y} > 0\}$, with the dual slacks $s_j = c_j - \boldsymbol{A}'_j \boldsymbol{\pi}$
- If $\boldsymbol{A}'_j \boldsymbol{y} \leq 0$ for every $j \notin \mathcal{J}$: the dual problem is unbounded and the primal problem is infeasible

Admissible set after the step:

- $j^\star$ enters, the basic variables of $RP(\mathcal{J})$ remain, only the indices with $\boldsymbol{A}'_j \boldsymbol{y} < 0$ may leave
- If $\boldsymbol{c} \geq 0$, the method can start from $\boldsymbol{\pi} = 0$

Pseudo-code of the primal-dual method

Introduction: Pseudo-code of the primal-dual method

- We put the pieces together: admissible set, restricted primal problem, dual step
- The pseudo-code repeats these three steps until the restricted primal problem has optimal value 0, or until the LP problem is shown to be infeasible
- Then we see why the dual objective value increases and why the method terminates

Pseudo-code (1/4)

Input
An LP problem in standard minimization form with $\boldsymbol{b} \geq 0$ and a feasible solution $\boldsymbol{\pi}$ of its dual problem.

Output
An optimal solution of the LP problem and an optimal solution of its dual problem, if they exist; otherwise, a certificate that the LP problem is infeasible.

Step 0: **Initialization**
Consider the given dual feasible solution $\boldsymbol{\pi}$. Go to Step 1.

Pseudo-code (2/4)

Step 1: **Admissible set**
Compute the dual slacks $s_j = c_j - \boldsymbol{A}'_j \boldsymbol{\pi}$ for every $j \in \{1, 2, \ldots, n\}$ and the admissible set $\mathcal{J} = \{ j : s_j = 0 \}$.

Step 2: **Restricted primal problem**
Solve $RP(\mathcal{J})$ with the simplex algorithm, obtaining its optimal value $\xi(\mathcal{J})$, an optimal solution x_j with $j \in \mathcal{J}$ and an optimal solution $\boldsymbol{y}$ of $DRP(\mathcal{J})$.

Pseudo-code (3/4)

Step 2: **Restricted primal problem** (continued)
If $\xi(\mathcal{J}) = 0$, then the point

$$\tilde{x}_j = x_j, \ \forall j \in \mathcal{J}, \qquad \tilde{x}_j = 0, \ \forall j \notin \mathcal{J},$$

is an optimal solution of the LP problem, $\boldsymbol{\pi}$ is an optimal solution of its dual problem, and $\boldsymbol{c}'\tilde{\boldsymbol{x}} = \boldsymbol{b}'\boldsymbol{\pi}$. `STOP`.

Pseudo-code (4/4)

Step 3: **Dual step**

a. If $\boldsymbol{A}_j'\boldsymbol{y} \leq 0$ for every $j \notin \mathcal{J}$, then the LP problem is **infeasible**. `STOP`.

b. Compute

$$\theta \leftarrow \min \left\{ \frac{s_j}{\boldsymbol{A}_j'\boldsymbol{y}} \ : \ j \notin \mathcal{J}, \ \boldsymbol{A}_j'\boldsymbol{y} > 0 \right\}.$$

c. Update the dual solution: $\boldsymbol{\pi} \leftarrow \boldsymbol{\pi} + \theta \, \boldsymbol{y}$.

Go to Step 1.

The dual objective value increases

- At every iteration the dual objective value $\boldsymbol{b}'\boldsymbol{\pi}$ strictly increases
- The finiteness of the method follows instead from the **restricted primal problems**:
  - the optimal basis of $RP(\mathcal{J})$ is still a feasible basis of the next restricted primal problem, since its indices remain admissible
  - the new admissible variable $x_{j^\star}$ has a negative reduced cost with respect to it

Finiteness of the method

- Accordingly, the next restricted primal problem can be solved starting from that basis
- The optimal values $\xi(\mathcal{J})$ never increase from one iteration to the next
- They strictly decrease if the pivoting operation on $x_{j^\star}$ is non-degenerate
- In general the method terminates after a finite number of iterations when an **anti-cycling rule** (for instance, Bland's rule) is used in the simplex algorithm applied to the restricted primal problems

Summary: pseudo-code of the primal-dual method

Steps:

- Step 0: start from a dual feasible solution $\boldsymbol{\pi}$
- Step 1: dual slacks and admissible set $\mathcal{J}$
- Step 2: solve $RP(\mathcal{J})$; if $\xi(\mathcal{J}) = 0$, the pair is optimal
- Step 3: infeasibility test, step length θ, update $\boldsymbol{\pi} \leftarrow \boldsymbol{\pi} + \theta\, \boldsymbol{y}$

Finiteness:

- $\boldsymbol{b}'\boldsymbol{\pi}$ strictly increases; $\xi(\mathcal{J})$ never increases, and an anti-cycling rule guarantees termination

Execution of the primal-dual method

Introduction: Execution of the primal-dual method

- We run the primal-dual method on an LP problem with 4 variables and 2 constraints
- At every iteration: dual slacks and admissible set, restricted primal problem, dual step
- We follow how the dual objective value grows and how the admissible set changes, until a primal feasible solution is found

The LP problem of the examples

We consider the following LP problem in standard minimization form with $n = 4$ non-negative variables and $m = 2$ equality linear constraints:

$$\begin{array}{rrrrrrrrrrrrrr} \min & 2 & x_1 & + & 2 & x_2 & + & 5 & x_3 & + & 3 & x_4 & & \\ \text{subject to} & & x_1 & + & & x_2 & + & 2 & x_3 & + & & x_4 & = & 6 \\ & & x_1 & + & 2 & x_2 & + & & x_3 & + & 3 & x_4 & = & 4 \\ & & x_1, & & & x_2, & & & x_3, & & & x_4 & \geq & 0 \end{array}$$

- Its right-hand side is non-negative and its objective-function vector is non-negative
- Then $\pi = (0 \;\; 0)'$ is a feasible solution of the dual problem

Let us now run the method from $\boldsymbol{\pi} = (0 \;\; 0)'$:
one iteration per slide.

Example: first iteration

Admissible set. With $\boldsymbol{\pi} = (0 \;\; 0)'$ the dual slacks are

$$s_1 = 2, \qquad s_2 = 2, \qquad s_3 = 5, \qquad s_4 = 3,$$

and then $\mathcal{J} = \emptyset$: no primal variable is allowed to be positive.

Restricted primal problem. Only the two artificial variables: $\xi(\emptyset) = a_1 + a_2 = 6 + 4 = 10 > 0$, and the optimal solution of $DRP(\emptyset)$ is $\boldsymbol{y} = (1 \;\; 1)'$.

Dual step. The four non-admissible columns give $\boldsymbol{A}_1'\boldsymbol{y} = 2$, $\boldsymbol{A}_2'\boldsymbol{y} = 3$, $\boldsymbol{A}_3'\boldsymbol{y} = 3$, $\boldsymbol{A}_4'\boldsymbol{y} = 4$, and

$$\theta = \min \left\{ \underbrace{\frac{2}{2}}_{=1}, \underbrace{\frac{2}{3}}_{=\frac{2}{3}}, \underbrace{\frac{5}{3}}_{=\frac{5}{3}}, \underbrace{\frac{3}{4}}_{=\frac{3}{4}} \right\} = \frac{2}{3}, \quad \text{attained by } j^\star = 2.$$

The new dual solution is $\boldsymbol{\pi} \leftarrow (0 \;\; 0)' + \frac{2}{3}(1 \;\; 1)' = \left(\frac{2}{3} \;\; \frac{2}{3}\right)'$, whose dual objective value is $\boldsymbol{b}'\boldsymbol{\pi} = \frac{20}{3}$.

Example: second iteration

Admissible set. With $\boldsymbol{\pi} = \begin{pmatrix} \frac{2}{3} & \frac{2}{3} \end{pmatrix}'$ the dual slacks are

$$s_1 = \frac{2}{3}, \qquad s_2 = 0, \qquad s_3 = 3, \qquad s_4 = \frac{1}{3},$$

and then $\mathcal{J} = \{2\}$.

Restricted primal problem. $RP(\{2\})$ has x_2 and the two artificial variables. Its optimal solution is $x_2 = 2$, with $\xi(\{2\}) = 4 > 0$, and the optimal solution of $DRP(\{2\})$ is $\boldsymbol{y} = \begin{pmatrix} 1 & -\frac{1}{2} \end{pmatrix}'$.

Dual step. For the non-admissible columns $\boldsymbol{A}_1'\boldsymbol{y} = \frac{1}{2}$, $\boldsymbol{A}_3'\boldsymbol{y} = \frac{3}{2}$, $\boldsymbol{A}_4'\boldsymbol{y} = -\frac{1}{2}$, and only the first two are positive:

$$\theta = \min \left\{ \underbrace{\frac{2/3}{1/2}}_{=\frac{4}{3}}, \underbrace{\frac{3}{3/2}}_{=2} \right\} = \frac{4}{3}, \quad \text{attained by } j^\star = 1.$$

The new dual solution is $\boldsymbol{\pi} \leftarrow \begin{pmatrix} \frac{2}{3} & \frac{2}{3} \end{pmatrix}' + \frac{4}{3} \begin{pmatrix} 1 & -\frac{1}{2} \end{pmatrix}' = (2 \;\; 0)'$, whose dual objective value is $\boldsymbol{b}'\boldsymbol{\pi} = 12$.

Example: third iteration

Admissible set. With $\boldsymbol{\pi} = (2 \quad 0)'$ the dual slacks are

$$s_1 = 0, \qquad s_2 = 0, \qquad s_3 = 1, \qquad s_4 = 1,$$

and then $\mathcal{J} = \{1,\ 2\}$.

Restricted primal problem. $RP(\{1,2\})$ has x_1, x_2 and the two artificial variables. Its optimal solution is $x_1 = 4$ and $x_2 = 0$, with $\xi(\{1,2\}) = 2 > 0$, and the optimal solution of $DRP(\{1,2\})$ is $\boldsymbol{y} = (1 \quad -1)'$.

Dual step. For the non-admissible columns $\boldsymbol{A}_3'\boldsymbol{y} = 1$ and $\boldsymbol{A}_4'\boldsymbol{y} = -2$, and then

$$\theta = \frac{s_3}{\boldsymbol{A}_3'\boldsymbol{y}} = \frac{1}{1} = 1, \quad \text{attained by } j^\star = 3.$$

The new dual solution is $\boldsymbol{\pi} \leftarrow (2 \quad 0)' + 1 \cdot (1 \quad -1)' = (3 \quad -1)'$, whose dual objective value is $\boldsymbol{b}'\boldsymbol{\pi} = 14$.

The second dual variable has become negative: this is allowed, since the dual variables are free.

Example: fourth iteration

Admissible set. With $\boldsymbol{\pi} = (3 \quad -1)'$ the dual slacks are

$$s_1 = 0, \qquad s_2 = 1, \qquad s_3 = 0, \qquad s_4 = 3,$$

and then $\mathcal{J} = \{1,\ 3\}$.

Restricted primal problem. $RP(\{1,3\})$ has x_1, x_3 and the two artificial variables. Its optimal solution is $x_1 = 2$ and $x_3 = 2$, with $\xi(\{1,3\}) = 0$: the method **stops**.

An optimal solution of this LP problem is

$$\tilde{\boldsymbol{x}}(\, LP\,) = \begin{pmatrix} 2 \\ 0 \\ 2 \\ 0 \end{pmatrix}$$

An optimal solution of its dual problem is $\tilde{\boldsymbol{\pi}}(\, D(LP)\,) = (3 \quad -1)'$, and the two optimal values coincide:

$$\boldsymbol{c}'\tilde{\boldsymbol{x}} = 2 \cdot 2 + 5 \cdot 2 = 14,$$
$$\boldsymbol{b}'\tilde{\boldsymbol{\pi}} = 6 \cdot 3 + 4 \cdot (-1) = 14,$$

in accordance with the Strong Duality theorem.

What happened along the iterations

- The dual objective value increased at every iteration,

$$0 \longrightarrow \frac{20}{3} \longrightarrow 12 \longrightarrow 14$$

- The admissible set changed accordingly,

$$\emptyset \longrightarrow \{2\} \longrightarrow \{1,2\} \longrightarrow \{1,3\},$$

  until it contained enough variables to build a primal feasible solution
- The admissible set does not necessarily grow: at the fourth iteration the index 2 left it, since $\boldsymbol{A}_2'\boldsymbol{y} = -1 < 0$ at the third iteration

Summary: execution of the primal-dual method

Four iterations from $\boldsymbol{\pi} = (0 \quad 0)'$:

- $\xi(\mathcal{J})$: $10 \to 4 \to 2 \to 0$
- $\boldsymbol{b}'\boldsymbol{\pi}$: $0 \to \frac{20}{3} \to 12 \to 14$
- $\mathcal{J}$: $\emptyset \to \{2\} \to \{1,2\} \to \{1,3\}$

Result:

- $\tilde{\boldsymbol{x}} = (2 \quad 0 \quad 2 \quad 0)'$ and $\tilde{\boldsymbol{\pi}} = (3 \quad -1)'$, both with value 14
- The dual variables may become negative, and indices may leave the admissible set

Comparison with the simplex algorithms

Introduction: Comparison with the simplex algorithms

- We now place the primal-dual method next to the simplex algorithm and the dual simplex algorithm
- What each method keeps, and what it works towards
- Why the primal-dual method is of practical interest, in particular for **combinatorial optimization** problems

Three methods

	keeps	**works towards**
simplex algorithm	a primal feasible basis	dual feasibility
dual simplex algorithm	a dual feasible basis	primal feasibility
primal-dual method	a dual feasible solution	primal feasibility

Basis versus solution

- The dual simplex algorithm keeps a dual feasible **basis**
- The primal-dual method keeps a dual feasible **solution**, which need not be basic
- For this reason the primal-dual method can start from any dual feasible solution, even one suggested by the structure of the problem

Practical interest of the primal-dual method

- The restricted primal problem is often much easier than the original LP problem: it has fewer variables
- For several **combinatorial optimization** problems it turns out to be a problem with a known combinatorial algorithm, such as a shortest path or a maximum flow problem
- In those cases the primal-dual method becomes a purely combinatorial algorithm, and it is the origin of many classical algorithms and of several approximation algorithms

Summary: comparison with the simplex algorithms

What each method keeps:

- Simplex: a primal feasible basis; dual simplex: a dual feasible basis
- Primal-dual: a dual feasible solution, not necessarily basic

Why it matters:

- The restricted primal problem has fewer variables and is often a combinatorial problem, such as a shortest path or a maximum flow problem

Summary (1/2)

- The primal-dual method works on an LP problem in standard minimization form with $\boldsymbol{b} \geq 0$ and keeps a **dual feasible solution** π
- The **admissible set** $\mathcal{J}(\pi)$ contains the indices of the dual constraints satisfied with equality: by Complementary Slackness, the pair is optimal if and only if only the admissible variables are positive
- The **restricted primal problem** $RP(\mathcal{J})$ looks for such a primal solution with m artificial variables: if $\xi(\mathcal{J}) = 0$, the method stops with an optimal pair

Summary (2/2)

- Otherwise, the optimal solution $\boldsymbol{y}$ of the **restricted dual problem** is an improving direction: $\boldsymbol{\pi} \leftarrow \boldsymbol{\pi} + \theta\, \boldsymbol{y}$, with the largest feasible step length θ
- If no non-admissible column has $\boldsymbol{A}'_j \boldsymbol{y} > 0$, the dual problem is unbounded and the LP problem is **infeasible**
- The dual objective value strictly increases, and with an anti-cycling rule the method terminates
- Unlike the dual simplex algorithm, the method keeps a dual feasible **solution**, not a basis: it is the origin of many combinatorial algorithms